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Machine Safety Specialist (CE)

Tab 1 – Module 3: Adjacent product Directives

TAB1: principles machine safety
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Product- and Social Directives

Examples of EU Product Directives	Examples of Social Directives
• Machine guidelines • EMC directive • Low voltage directive • Pressure equipment directive • ATEX directive	• Directive Work Equipment Directive • Directive for Manual Handling of Loads • ATEX Directive for Employers • Directive on Display Screen Equipment
In the Netherlands adopted in the "Commodities Act".	In the Netherlands adopted in the Occupational Health and Safety Legislation.

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Low Voltage Directive 2014/35/EU

- Previously 2006/95/EG and 73/23/EEG.
- Incorporated in the Commodities Act Decree on Electrotechnical Products.

Low voltage directive 2014/35/EU

Requirements for electrical equipment:

~ Between 50 V and 1000 V
= Between 75 V and 1500 V

The Directive requires:

- *Technical Documentation*
- *Internal manufacturing control*
- *EU-Declaration of Conformity*

The Machine Directive covers requirements for electrical installation of machinery (Low Voltage Directive is applicable, but not on EC Declaration of Conformity for machines).

Important standard: EN-IEC 60204-1 – Electrical equipment of machines

Guide

1.5 RISKS DUE TO OTHER HAZARDS

1.5.1 Electricity supply

Where machinery has an electricity supply, it must be designed, constructed and equipped in such a way that all hazards of an electrical nature are or can be prevented.

The safety objectives set out in Directive 73/23/EEC shall apply to machinery. However, the obligations concerning conformity assessment and the placing on the market and/or putting into service of machinery with regard to electrical hazards are governed solely by this Directive.

The first paragraph of section 1.5.1 requires the machinery manufacturer to take the necessary measures to prevent all hazards of an electrical nature. This general requirement always applies, irrespective of the voltage of the electrical supply.

The second paragraph of section 1.5.1 declares the safety requirements of the Low Voltage Directive 2006/95/EC (previously 73/23/EEC as amended) applicable to machinery. The second sentence of this paragraph indicates that the procedures of the Low Voltage Directive relating to the placing on the market and putting into service are not applicable to machinery subject to the Machinery Directive. This means that the Declaration of Conformity for machinery subject to the Machinery Directive may not make reference to the Low Voltage Directive.

EMC-Directive 2014/30/EU

- Previously 2004/108/EG
- Incorporated in the Legislation on Telecom Facilities

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EMC-Directive 2014/30/EU

Appliances, equipment: all electrical and electronic appliances, as well as equipment and installations containing electrical and/or electronic components;

Equipment must, taking account of the state of the art, be designed and manufactured in such a way as to guarantee that:

a) the electromagnetic disturbance generated does not exceed a level beyond which radio and telecommunications equipment and other equipment cannot operate as intended;

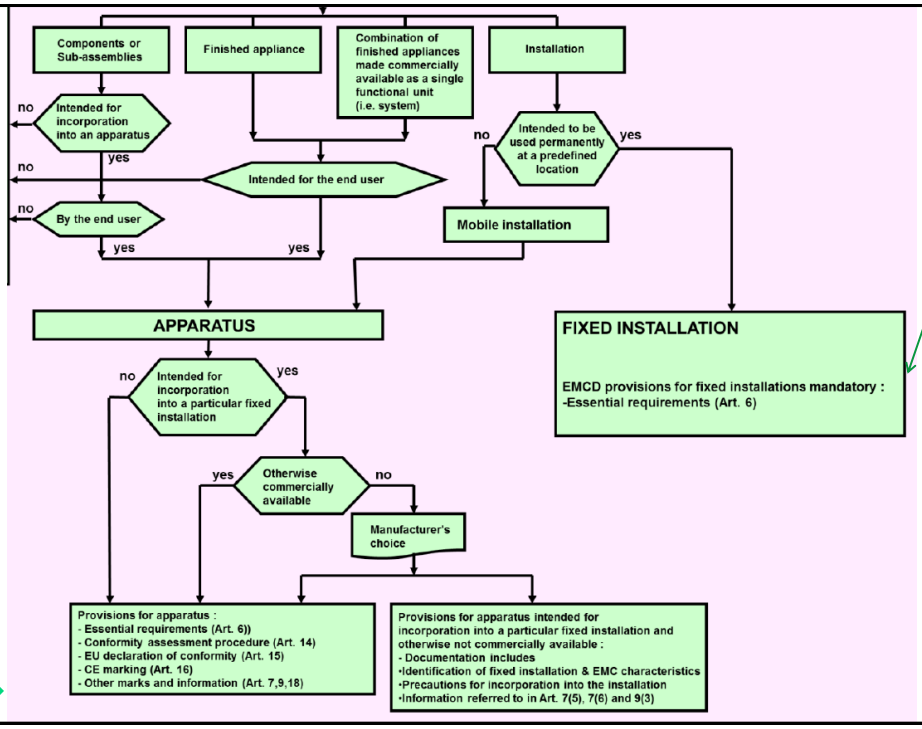
b) it shall have a level of immunity to the electromagnetic disturbance to be expected in its normal application which allows it to operate without unacceptable degradation of its intended use.

Emission

&

Immunity

Tab1-3/5-Sh.8



„Fixed installation": a particular combination of several types of apparatus and, where applicable, other devices, which are assembled, installed and intended to be used permanently at a predefined location.;

Flowchart from the guide for EMC Directive

Pressure Equipment Directive 2014/68/EU

- Previously 97/23/EG
- Incorporated in the Commodities act Decree Pressure Equipment

Pressure Equipment Directive

Scope:
- Pressure vessels,
- Process piping,
- Safety accessories,
- Pressure accessories,
- Assemblies,
- (Fire) Heated pressure equipment,

With a design pressure greater than 0,5 bar.

Categories I up to IV, with 'good craftmanship' as residual category.
Exemptions listed in article 1(2).

ATEX 114 Directive 2014/34/EU

- Previously 94/9/EG (ATEX 95).
- CE-marking and EX-marking
- Incorporated in the Commodities Act Decree on Explosion safe Equipment.

ATEX 114 Directive

Applies to equipment and protective systems intended for use in **potentially explosive atmospheres.**

Article 1 (2)
- machines,
- equipment,
- fixed or mobile devices,
- control devices and instruments,
- detection and prevention systems,

which, alone or in combination, are intended for production, transport, storage, measurement, control, energy conversion and/or processing of raw materials and which can cause an explosion through their inherent **potential sources of ignition.**

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Directives

Downloadable from: http://eur-lex.europa.eu

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